

Bitana V6 — Complete System Audit & Evolution Report

Document Version: 2.0

Date: 2026-05-30

Author: OWL (for Martin)

Repository: `github.com:vsehuinya1/bitana`

Branch: `v6.4.1-hotfix` (8 commits ahead of main)

Latest Commit: `de88376` — BD_FILTER log level fix

Table of Contents

1. [Executive Summary](#)
 2. [What Changed Since v1.0 \(May 25\)](#)
 3. [System Architecture Overview](#)
 4. [Hotfix Changes \(v6.4 → v6.4.1\)](#)
 5. [Post-Gate Performance](#)
 6. [Trajectory Analysis & Regime Discovery](#)
 7. [State Transition Logic](#)
 8. [MAE Persistence Kill Rule](#)
 9. [Pathway Expectancy Analysis](#)
 10. [Telemetry Integrity Audit](#)
 11. [Proven vs Experimental Pairs](#)
 12. [Open Issues & Next Steps](#)
 13. [Current Configuration Reference](#)
 14. [Full Git Log](#)
-

1. Executive Summary

Bitana is a liquidation-cascade breakout trading system for Binance USDT-M perpetual futures, executing on 5m candles. It evolved from V3 through v6.4.1-hotfix over approximately 3 weeks of intensive development.

Current state (v6.4.1-hotfix, May 30): - 255 all-time closed trades, 43% WR, -51.93R total (includes pre-gate) - **Post-gate (63 trades): 47.6% WR, +1.46R total, expectancy +0.023R/**

trade ← barely positive - 57 symbols (35 proven + 22 experimental) - 14/57 cascades currently active - System is running with BD < -2% gate active (deployed May 27)

Critical findings from v6.1.5 analysis: 1. **The system has 5 latent trajectory regimes, not a continuum** — 2 winner types (EXPLOSIVE convex runners + SURVIVOR slow grinders), 3 loser types (immediate rejection, flat chop, early spike fade) 2. **London session is the primary bleed source** — 20% WR, -6.21R over 15 trades. Skipping London = +7.67R, 56% WR across remaining sessions 3. **All 30 post-gate winners exited via vol_trail** — exit reason consistency is 100% 4. **The imb_z gate remains broken** — confirmations field shows all gates passing on every trade (data recording issue, not gate logic) 5. **r_path telemetry is fully operational** — 26,398 rows across 151 trades, 100% coverage of post-gate trades 6. **MAE recovery is the #1 winner/loser separator** (Cohen's $d=1.94$) — winners recover 1.6R from trough, losers only 0.42R

Path to profitability: Either fix London session (skip or add regime filter) OR reduce average loss from -0.92R to -0.50R via kill rules. Both paths get expectancy to +0.15-0.25R/trade.

2. What Changed Since v1.0

Date	Change	Impact
May 25	v1.0 report published	Baseline: 149 trades, -28.25R, -0.19R expectancy
May 25	Hotfix branch created (v6.4.1)	Started surgical fixes
May 26	imb gate fix (synthetic fallback removed)	imb_z now returns 0.0 when data missing
May 26	BD < -2% gate deployed	Blocks structurally bad entries
May 26	min_confirmations 3 → 4	Reverted V6.4 loosening
May 27	BD gate goes live	837+ entries blocked in first days
May 28-30	Trajectory analysis + tensor construction	Full regime discovery (Phases 1.5-2)
May 30	Pathway expectancy analysis	8/13 pathways show positive expectancy

3. System Architecture Overview

Strategy: 3-Play Liquidation Cascade Breakout

Play	Description
A) Breakout Continuation	Enters when price breaks above 60-bar range high with cascade backing
B) Liq Cascade Capture	Enters when liquidation cluster is detected and cascade strength exceeds threshold
C) Momentum Ignition	Enters when reclaim + momentum signals align after cascade activates

Confirmation Stack (6 gates)

Gate	Current Threshold	Notes
body	> 0.50	% of candle body to range
imb_z	> 1.0	Taker buy/sell volume z-score (BUG: returns 0.0 when data missing)
vol_z	> 2.0	Volume z-score
impulse	> 0.20	Impulse strength
breakout	above range_high	Price breaks 60-bar range (NEVER fires — inversely predictive)
momentum	positive	Momentum direction

min_confirmations: 4 (reverted from V6.4's 3)

Exit Logic (per-decile parameters)

All winners exit via **vol_trail** (ATR-based trailing stop). Losers exit via **stop_loss** (full -1R) or **early_cut** (decay detection).

Key exit params vary by decile: **vol_trail_atr** (1.5-3.0), **max_holdBars** (100-500), **structLookback** (8-48).

4. Hotfix Changes (v6.4 → v6.4.1)

Change A: imb Gate Fix

File: `engines/liq_cluster_engine_v5.py` **Problem:** When `taker_buy_volume` was zero/invalid, the imb calculation used a synthetic fallback that produced misleadingly high `imb_z` values **Fix:** When taker data is missing/invalid, `imb_z` now correctly returns 0.0 **Impact:** imb confirmation now properly fails when data is unavailable

Change B: BD Lower-Bound Gate

File: `engines/liq_cluster_engine_v5.py` **Logic:** `breakout_distance_pct = (close - max(high[-61:-1])) / max(high[-61:-1]) * 100` **Threshold:** Reject entry if `breakout_distance_pct < -2.0%` **Rationale:** BD < -2% was the worst loss cluster: 25 trades, 32% WR, -13.40R pre-gate **Status:** Deployed and active since 2026-05-27T21:00 UTC

Change C: min_confirmations 3 → 4

Rationale: V6.4 loosened confirmations (4 → 3). Result: 18 trades, -11.16R, 10.5% WR. Audit showed all 18 trades would have been blocked at 4. Reverted.

Change D: Breakout Distance Logging

Added `range_high`, `breakout_distance_pct`, `imb_z`, `imb_fallback_triggered` to `signal_data` for runtime visibility.

5. Post-Gate Performance

Overall (post-gate, 63 trades)

Metric	Value
Trades	63 (30W / 33L)
Win Rate	47.6%
Total R	+1.46R
Winner R	+31.93R (avg +1.064R)
Loser R	-30.47R (avg -0.923R)
Expectancy/trade	+0.023R

Metric	Value
Kelly fraction	2.2%

Winner Exit Reasons

Reason	Count	Total R	Avg R
vol_trail	30/30	+31.93R	+1.064R

100% of winners exit via vol_trail. Zero exceptions.

Loser Exit Reasons

Reason	Count	Total R	Avg R
stop_loss	29/33	-29.72R	-1.025R
vol_trail	3/33	-0.14R	-0.045R
early_cut	1/33	-0.62R	-0.617R

Session Breakdown (post-gate)

Session	WR	Total R	Verdict
Asia (00-08 UTC)	54%	+3.16R	✓ Profitable
London (08-14 UTC)	20%	-6.21R	✗ Primary bleed
NY (14-21 UTC)	69%	+5.81R	✓ Profitable
LateNY (21-24 UTC)	33%	-1.29R	⚠ Slight negative

Winners by Size

Size	Count	Total R	Avg R
<0.5R	7	+1.94R	+0.28R
0.5-1R	11	+8.46R	+0.77R
1-2R	10	+14.44R	+1.44R
3R+	2	+7.09R	+3.55R

Recent examples from live Telegram: - INJ +3.77R (vol_trail, 45c) - ETH +1.58R (vol_trail, 34c) - DOT +1.24R (vol_trail, 11c) - VIRTUAL +0.52R (vol_trail, 29c)

Daily Report (May 29)

- 26 trades, 50% WR, +4.12R
- All-time: 255 trades, 43% WR, -51.93R
- Equity: \$680.34 (DD 93.2% from peak — includes pre-gate losses)

6. Trajectory Analysis & Regime Discovery

Based on 151 trades with full `r_path` data, trajectory clustering identified **7 regimes** that collapse into **3 actionable classes**.

3 Actionable Classes

Class	N	WR	Avg PnL	Avg MFE	Avg Bars	Exit
EXPLOSIVE (convex runners)	26	100%	+0.91R	1.89R	29	All <code>vol_trail</code>
SURVIVOR (slow grinders)	26	100%	+0.43R	0.74R	365	Mixed
DEAD (all losers)	99	0%	-0.81R	0.19R	223	Mostly <code>stop_loss</code>

EXPLOSIVE: MFE > 1.0R within 15 bars, positive acceleration. Short hold, strong expansion. SURVIVOR: Winners with either MFE < 1.0R OR > 20 bars to reach 1R. Long hold, low MAE, modest MFE. DEAD: Everything else. Subclassified as EARLY_DEAD (MFE never > 0.3R in first 10 bars), LATE_DEAD (MFE > 0.3R but still lost), FADE_DEAD (MFE > 0.5R then collapsed).

Winner vs Loser: Cold Facts (Cohen's d)

Feature	Winners	Losers	d	Separation
Max MAE Recovery	1.60R	0.42R	+1.94	★ Best separator
Max MFE	1.32R	0.21R	+1.69	★ Strong
MFE Velocity (late)	+0.060	-0.002	+0.86	★ Strong
Stale % (deep MAE + no growth)	22%	50%	-0.79	Strong
Max MAE Depth	0.28R	0.59R	-1.06	Strong
Deep Bleed %	26%	51%	-0.67	Moderate

Feature	Winners	Losers	d	Separation
MFE Velocity (early)	0.049	0.020	+0.61	Moderate
Consec Deep Bleed	122b	119b	+0.01	✗ No separation

Time-to-First-Expansion Distribution

Threshold	Winners median	Losers median	Never reached
MFE $\geq$ 0.10R	3 bars	3 bars	W: 1/52, L: 51/99
MFE $\geq$ 0.20R	4 bars	3 bars	W: 7/52, L: 58/99
MFE $\geq$ 0.30R	5 bars	4 bars	W: 8/52, L: 67/99
MFE $\geq$ 0.50R	8 bars	5 bars	W: 11/52, L: 83/99

The regime boundary is at 0.30R, not 0.20R. Below 0.30R, winners and losers look identical in timing. At 0.30R+, they diverge sharply: 67% of losers never reach it vs 15% of winners.

7. State Transition Logic

State machine traced across 151 trades with 26,398 bar observations.

Transition Frequencies

From	To	Count
ENTRY	IGNITION	151
IGNITION	EXPANSION	88
IGNITION	CONFIRMATION	37
IGNITION	DECAY	19
EXPANSION	DECAY	63
EXPANSION	CONFIRMATION	43
CONFIRMATION	EXPANSION	23
CONFIRMATION	DECAY	2
DECAY	EXPANSION	26
DECAY	CONFIRMATION	8

State Durations

State	Median	Avg
IGNITION	3 bars	3 bars
EXPANSION	4 bars	42 bars
CONFIRMATION	8 bars	24 bars
DECAY	22 bars	439 bars

Exits by Final State

Final State	Exit Reasons
EXPANSION	7 early_cut, 7 time_stop, 14 stop_loss, 3 struct_trail
CONFIRMATION	41 vol_trail, 4 struct_trail, 1 cascade_deactivated, 17 stop_loss
DECAY	39 stop_loss, 2 time_stop, 8 early_cut, 1 vol_trail

Key insight: 41 of 41 vol_trail exits came from CONFIRMATION state. The path to winning: get to CONFIRMATION (MFE > 0.3R while still expanding). Only 2 trades exited from CONFIRMATION via early_cut — it's a safe state.

DECAY is where trades go to die: 39 stop_loss exits, median 22 bars in the state.

8. MAE Persistence Kill Rule

Rule Definition

```
IF bar >= 5
  AND MAE > 0.5R
  AND MFE velocity < 0.01 for 3+ consecutive bars
  AND MAE recovery < 0.3R
THEN kill at market (est. -0.50R)
```

Results (in-sample, 151 trades)

Metric	Value
True kills (losers caught)	37/99 (37%)
False kills (winners killed)	10/52 (19%)

Metric	Value
Missed losers	62/99 (63%)
Clean winners	42/52 (81%)

By Actionable Class

Class	True Kills	False Kills	Clean
EXPLOSIVE	0	1	25
SURVIVOR	0	9	17
EARLY_DEAD	32	0	40
LATE_DEAD	3	0	11
FADE_DEAD	2	0	11

False Kills (winners destroyed)

Symbol	PnL	Class	Killed at bar	Would have been
INJ	+0.87R	SURVIVOR	7/57	vol_trail
AAVE	+0.78R	SURVIVOR	844/3086	time_stop
NEAR	+0.76R	SURVIVOR	12/39	vol_trail
SOL	+0.74R	SURVIVOR	146/2411	time_stop
QNT	+0.41R	SURVIVOR	220/343	struct_trail
ZEC	+0.37R	EXPLOSIVE	13/70	vol_trail
NMR	+0.28R	SURVIVOR	17/47	vol_trail
INJ	+0.09R	SURVIVOR	15/26	vol_trail
LINK	+0.08R	SURVIVOR	7/151	struct_trail
LTC	+0.07R	SURVIVOR	25/36	vol_trail

Expectancy Impact

Metric	Original	With Rule	Delta
Expectancy/trade	-0.309R	-0.283R	+0.026R
Total R (151 trades)	-46.58	-42.71	+3.88R

Metric	Original	With Rule	Delta
Compounding @ 0.5% risk	\$79.10	\$80.67	+\$1.57

Assessment

The rule marginally improves expectancy (+0.026R/trade) but is too blunt. It catches 37% of losers by cutting them at -0.50R instead of -1.00R, but kills 19% of winners — mostly SURVIVOR class trades that dip deep then recover over long timeframes. The core failure mode: confusing "temporary adversity" with "non-viability."

Not recommended for implementation as-is. The SURVIVOR false kills are exactly the trades you want to keep.

9. Pathway Expectancy Analysis

13 pathway filters tested. 8 show positive expectancy.

Positive Pathways

Pathway	N	WR	Exp/R	Kelly	\$100 → (1% risk)	MaxDD
EXPANSIVE (MFE>1.5R in 15 bars)	16	100%	+1.31R	—	\$123 (+23%)	0%
FAST WINNER (hold < 20 bars)	11	100%	+0.71R	—	\$108 (+8%)	0%
CONFIRMED + HIGH RECOVERY	35	86%	+0.63R	72%	\$124 (+24%)	2%
HIGH RECOVERY (>1R from trough)	50	88%	+0.59R	76%	\$134 (+34%)	1%
SUSTAINED ACCEL (vel>0 late)	49	82%	+0.46R	60%	\$125 (+25%)	3%
CONFIRMED + SHALLOW MAE	40	68%	+0.36R	39%	\$115 (+15%)	5%
SHALLOW MAE (<0.4R)	56	62%	+0.19R	27%	\$111 (+11%)	9%
CONFIRMED (MFE>0.3R in 10 bars)	57	54%	+0.06R	7%	\$103 (+3%)	10%

Most Practical: HIGH RECOVERY

- 50 trades, 88% WR, +0.59R expectancy

- At 1% risk: \$100 → \$134 (+34%), MaxDD only 1%
- This is the "don't kill trades that recover from troughs" principle quantified

Critical Caveat

Most positive pathways are post-hoc. You only know "HIGH RECOVERY" after the trade has recovered. The one fully real-time filter is "CONFIRMED (MFE>0.3R in 10 bars)" at +0.06R expectancy — barely positive.

The real-time question: can you build a classifier that identifies EXPLOSIVE vs DEAD before bar 10?

10. Telemetry Integrity Audit

r_path Status

Metric	Value
Rows	26,398
Trades covered	151
Post-gate coverage	56/56 (100%)
Per-bar fields	13 (price, unrealized_r, mae_so_far, mfe_so_far, vol_trail_level, struct_trail_level, atr, consecutive_red, above_ema, above_range_high)

MFE Tracking Accuracy

Check	Result
vol_trail exits with MFE=0	✅ Zero (all track properly)
early_cut exits with MFE=0	✅ By design (condition is mfe < 0.1R)
r_path max_mfe vs trades.mfe	⚠️ 1-bar lag on some trades

The 1-bar lag: Runner logs r_path BEFORE calling manage_position(). On bar 1, engine MFE is 0 (hasn't run yet). So r_path MFE at bar N = engine MFE at bar N-1. On the close bar, the close result overwrites p["mfe"], so trades.mfe is always correct. r_path is off by at most one bar's increment.

MFE/MAE Correctness on Key Trades

Trade	pnl_r	trades.mfe	r_path max	Match?
ETH +1.73R	+1.73	2.599	2.599	✓
LTC +1.56R	+1.56	2.165	2.165	✓
HYPE +1.24R	+1.24	2.342	1.859	⚠ (-0.48R)
QNT +0.94R	+0.94	1.825	0.944	⚠ (-0.88R)

early_cut Bug

early_cut fires for ALL deciles (D1-D10) regardless of `decay_enabled` setting. For D1/D2/D3 (decay_start_bar=999), it only fires at bar 999+, which means it fires AFTER time_stop (max_hold_bars=288-500) would have already closed the trade. In practice, early_cut fires for D9/D10 (decay_start_bar=8) and for deciles where a trade survives long enough via time_stop to reach bar 999+.

Confirmation Recording Bug

All 63 post-gate trades show all 5 confirmations (body, imb_z, vol_z, impulse, breakout, momentum) as passing — on BOTH winners and losers. This is a **data recording issue**, not gate behavior. The confirmations field isn't being updated post-hotfix. Needs investigation.

11. Proven vs Experimental Pairs

All-time (255 trades)

	Proven (169t)	Experimental (86t)
WR	47%	35%
Total R	-34.90R	-17.03R
Exp/trade	-0.207R	-0.198R

Both are negative. Experimental is slightly less negative per trade.

Last 7 Days Post-Gate

Date	Proven	Experimental
May 27	-4.68R	+1.50R

Date	Proven	Experimental
May 28	-3.88R	+1.34R
May 29	-0.69R	+6.20R
May 30	+0.73R	-2.12R

Experimental carried May 29 (8/15 trades, +6.20R).

The Decile Split

Decile	Proven	Experimental
D1	89t, -20.34R	54t, -20.40R
D2	26t, -6.53R	13t, +4.54R
D6	9t, -2.25R	4t, +2.41R

D2 experimental is +4.54R while D2 proven is -6.53R. This is what's driving the difference.

Experimental Top Symbols

Symbol	N	WR	Total R
INJUSDT	11	55%	+4.01R
VIRTUALUSDT	3	67%	+1.44R
FILUSDT	5	40%	+0.86R
AVAXUSDT	9	11%	-6.41R
HYPEUSDT	9	22%	-4.27R

INJ alone is +4.01R. Without INJ, experimental is flat or negative. The outperformance is NOT systematic — it's 1-2 symbols carrying the group.

Recommendation

Do NOT expand to more experimental pairs blindly. The edge is symbol-specific (INJ), not a category effect. Adding more random pairs will likely dilute the edge.

If expanding: target pairs that behave like D2 experimental (where the +4.54R came from). Need to identify what makes D2 experimental different from D2 proven — symbol selection, liquidity profile, or session behavior.

12. Open Issues & Next Steps

Critical Issues

1. **Confirmation recording broken** — all gates show pass on every trade post-hotfix.
Need to fix the confirmation logging in the runner.
2. **imb_z gate data quality** — returns 0.0 when taker data missing, but the confirmations field doesn't record this accurately.
3. **London session bleed** — 20% WR, -6.21R. Either skip London entirely or build a regime filter.

High-Priority Analysis

1. **Build real-time classifier** — Can you identify EXPLOSIVE vs DEAD before bar 10 using only information available at that bar? This is the highest-leverage question.
2. **Composer implementation** — The tensor and regime analysis is ready for Composer to build enforceable state-machine logic, instrumentation, and forward-testing pipelines.
3. **Validate CONFIRMED pathway** — MFE > 0.3R within 10 bars is the only real-time filter with positive expectancy (+0.062R). Test out-of-sample on next 30-50 trades.

Forward Test Specification

Single rule to test: MAE-Recovery Kill - Rule: $MAE > 0.5R$ + MFE velocity < 0.01 for 3+ bars + recovery < 0.3R, after bar 5 - Abort threshold: >25% false kill rate on winners in forward sample - Success: forward expectancy > 0 AND WR > 45% - Sample needed: 30+ trades for statistical validity

But: In-sample results show +0.026R expectancy improvement with 19% false kill rate on winners. This is marginal. Composer should refine the rule before deployment.

13. Current Configuration Reference

V5Config (current on VPS, v6.4.1-hotfix)

Parameter	Value
min_confirmations	4
vol_z threshold	2.0
imb_z threshold	1.0

Parameter	Value
body threshold	0.50
impulse threshold	0.20
BD gate	REJECT if BD < -2.0%
liq_lookback	90 days
p90 cascade filter	Active (sole cascade filter)
min_cascade_imb	DISABLED

DECILE_EXITS (key params)

Decile	vol_trail_atr	struct_lookback	max_holdBars	decay_enabled	decay_start
D1	3.0	48	500	No	999
D2	3.0	48	500	No	999
D3	2.0	24	288	No	999
D4	2.0	12	288	Yes	15
D5	2.0	12	288	Yes	15
D6	2.0	12	288	Yes	12
D7	2.5	36	358	Yes	20
D8	2.5	36	358	Yes	20
D9	1.5	8	100	Yes	8
D10	1.5	8	100	Yes	8

Service Info

- Service: `bitana-v5-paper.service`
 - Branch: `v6.4.1-hotfix`
 - DB: `storage/v5_forward_test.db` (trades), `storage/v6_telemetry.db` (r_path)
 - Runner: `tools/v5_forward_test.py` (1,403 lines, unmodified)
 - Engine: `engines/liq_cluster_engine_v5.py` (hotfixes applied)
-

14. Full Git Log

```
de88376 fix: change BD_FILTER log level from debug to info
174fe56 v6.4.1: add BD lower-bound gate – reject entries with BD < -2.0%
a8180fc v6.4.1: revert min_confirmations 3→4
df5b661 v6.4.1: change imb_gate and breakout_gate logs from debug to info
e5d2601 v6.4.1-hotfix: fix imb gate, add breakout logging, consolidate decile filter
6d4a6a6 docs: add V6 audit report v1.0
456dbed v6.4: loosen entry confirmations (V6.4 loosening – later reverted)
b2bfabf feat: disable min_cascade_imb gate
d8864fc revert: restore liq_lookback to 90 days
1842eef fix(forward-test): align VPS recovery checks, shorten liq lookback
c4550ee fix: exit timestamp -5m offset bug
971bf2d widen early_cut threshold to R < -0.5 and MFE < 0.1
ccefbda fix(forward-test): resolve candle loop double-processing bug
8502cd9 add 7 new experimental pairs
a38de21 surgical experimental-pair tracking
d91b670 feat(telemetry): implement async bounded queue isolation layer
e81b043 feat(research): add analyze_telemetry.py
2987ce9 feat(telemetry): deploy v6.2 research telemetry, shadow exits
cb03ba7 V6.0: Full audit fix – 18 bugs fixed
...
```

End of report v2.0. Generated 2026-05-30. For the latest data, check the `r_path` table in `v6_telemetry.db` and the analysis scripts in `tools/`.